2.2 Types of Reasoning

Lesson Introduction

Benchmark	MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Targets	• I can write a conjecture. • I can write a counterexample. • I can use inductive and deductive reasoning.

	Building On	Working Toward
Benchmark Coherence	MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.	N/A

Essential Questions	• How can I write a conjecture? • How can I use the information provided to write a counterexample? • What are the differences in using inductive and deductive reasoning?
Lesson Narrative	In this lesson, students are introduced to different types of reasoning. Students will begin by exploring types of reasoning, while being introduced to the formal definitions of a conjecture, deductive reasoning, and inductive reasoning. They will discover how definitions, postulates, properties, and theorems can play a role in drawing an appropriate conclusion. Students will also be introduced to the concept of a counterexample.

Component Summary	2.2.1 Warm-Up (~5 min.)	Determining a pattern activity *(SE p. 89)*	2.2.3 Bring It Together (10 min)	Identifying types of reasoning and counterexamples *(SE p. 93)*
	2.2.2 Exploration (30 min)	Exploring different types of reasoning *(SE p. 90)*	2.2.4 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 93)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Conjecture • Deductive Reasoning • Inductive Reasoning • Counterexample <u>Additional Terms:</u> N/A	N/A	• Students may need more examples of statements that represent inductive and deductive reasoning. Have additional examples prepared if these are needed.

Teacher Tips

Common Misconceptions	Things to Consider
• Students may confuse deductive reasoning with inductive reasoning. • Students may be able to apply the concepts to real-world scenarios but not apply to mathematical concepts. • Students may inadvertently write a converse statement instead of a counterexample.	• This lesson is language- and vocabulary-heavy. It may be helpful for students to get extra practice in both real-world examples and mathematical scenarios.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Think Pair Share In the 2.2.2 Exploration activity, students will work together to make conjectures. Students will have the opportunity, in questions 1 and 2, to make a conjecture on their own and then work with their partner to share their answers while summarizing their reasons.	**MA.K12.MTR.4.1: Discussion and Reflection** 2.2.2 Exploration provides students with an opportunity to work with a partner to engage in active discussion and reflection based on the prompts and instructions provided in the activity. Teachers should look for opportunities to develop students' ability to justify methods. Students may need support from sentence stems to understand how to respond to their peers.

Differentiate Instruction

Using the flow charts for deductive and inductive reasonings, consider creating a flow chart with examples of definitions, postulates, properties, theorems, a pattern, and a conjecture so visual learners can connect the process for future reference.

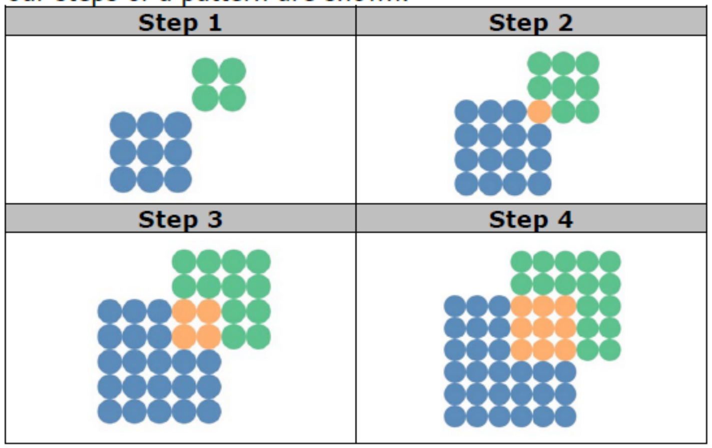
Lesson Notes:

2.2.1 Warm-Up: What's the Pattern?

Component Overview: *Given the first four steps of a pattern with different colored circles, the students will anticipate the number of circles in the fifth step of the pattern. For question 4, students will describe the pattern observed.*

2.2.1 Warm-Up: What's the Pattern?

Four steps of a pattern are shown.



1. How many blue circles will be in step 5?
   33 blue circles
2. How many orange circles will be in step 5?
   16 orange circles
3. How many green circles will be in step 5?
   20 green circles
4. Describe the pattern using words or an algebraic expression.
   Answers will vary. Blue circles follow the pattern of $6n + 3$, where n is the step number in the pattern. Orange circles follow the pattern of $(n - 1)^2$, where n is the step number in the pattern. Green circles follow the pattern of $4n$, where n is the step number in the pattern.

If students do not write an algebraic expression for the pattern, avoid giving hints for doing this, as they will be writing this in the next component, 2.2.2 Exploration.

2.2.2 Exploration: Types of Reasoning

Component Overview: *Students will work together to complete an Exploration on different types of reasoning.*

2.2.2 Exploration: Types of Reasoning

To describe a pattern such as the pattern in the Warm-Up is to make a conjecture.

A ***conjecture*** is a mathematical statement that has not yet been proven.

Once a conjecture has been made, then reasoning can be used to prove the conjecture.

1. Work with your partner to write conjectures for the pattern in the Warm-Up. You should be able to use each conjecture to determine any step in the pattern.

   - number of blue circles
   - number of orange circles
   - number of green circles
   - total number of circles

Blue	Orange	Green	Total
$6n+3$ is the equation to find the number of blue circles, where n is the step number.	*$(n-1)^2$ is the equation to find the number of orange circles, where n is the step number.*	*$4n$ is the equation to find the number of green circles, where n is the step number.*	*n^2+8n+4 is the equation to find the total number of circles, where n is the step number.*

A conjecture is an important step in problem-solving. Conjectures arise when you notice a pattern and that pattern holds true for many cases. However, just because a pattern holds true for many cases does not mean that the pattern will hold true for ***all*** cases. Conjectures must be proven for the mathematical observation to be fully accepted. When a conjecture is rigorously proven, it becomes a theorem.

Students may not be familiar with the word "conjecture." If your students need further support consider beginning this lesson by introducing the concept of conjecture and release students to complete the activity up through the flow chart for deductive and inductive reasoning. Once you introduce those concepts, rerelease students to complete the remainder of the activity.

While students are working with their partner, monitor that students are moving forward with completing the table. The focus of the lesson is on developing a conjecture, not the algebra of writing the expression. Prepare questions prior to class that may assist them if they struggle with the table. Sample questions are:

- *Can you create a table for each color with the step number as the input and the number of circles as the output? What patterns do you notice for the blue and green circles? What type of numbers are the numbers of orange circles?*
- *How can you write a pattern using the step number as your input (variable)?*

2.2.2 Exploration: Types of Reasoning (continued)

The values for step 10 are shown. Use these values to test your conjecture. Revise your conjecture if necessary.

Blue	Orange	Green	Total
63	81	40	184

2. Ask a classmate for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the conjecture and summary columns of the table.* Have the person initial next to your summary stating that your summary was correct.
   Answers will vary.

Color	Conjecture	My Summary of Their Reason	Initials
Blue			
Orange			
Green			
Total			

The conjectures from your classmate may be different from your conjectures, but that does not mean that one is wrong. Both may be correct.

3. Compare your conjecture with the conjectures you and your partner collected from different classmates.

4. The values for step 18 are shown. Use these values to test your classmates' conjectures.

Blue	Orange	Green	Total
111	289	72	472

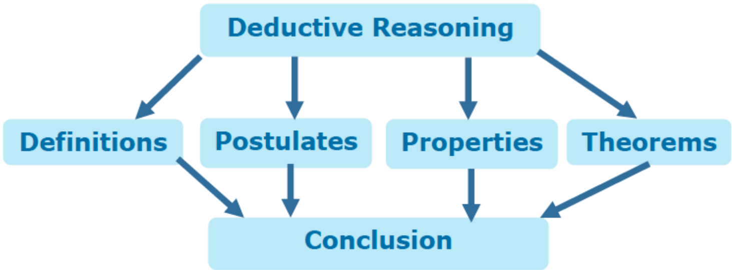
Answers will vary.

Students may need directions for question 2 in terms of logistics and a reminder of classroom expectations for their level of movement and conversation. If classroom behavior is an issue, consider assigning each student a specific partner within an appropriate proximity to avoid classroom disruption during this piece. The classmate for question 2 is not their original partner for the 2.2.2 Exploration.

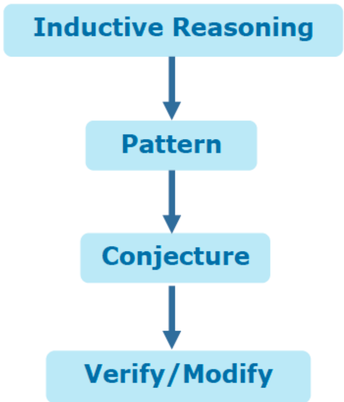
2.2.2 Exploration: Types of Reasoning (continued)

There are two types of reasoning that can be used to prove a conjecture.

Deductive reasoning uses facts and general principles to draw conclusions.



Inductive reasoning uses specific observations and patterns to draw conclusions.



Consider creating an anchor chart for this step that helps students have a visual reference when dealing with deductive and inductive reasoning.
For a note-taking reference guide, consider providing examples where things they learned in prior mathematics would be used as a form of deductive reasoning, such as the distributive property or triangle sum theorem.

5. Discuss with your partner which type of reasoning best describes the type that was used in the warm-up. Summarize your discussion.
Answers will vary. The patterns are very much an inductive reasoning process, so the warm-up must be an example of inductive reasoning.

Much of the reasoning that will be used in geometry is deductive reasoning.

Deductive reasoning needs to be used to prove the following statement is true.

All numbers ending in 0 or 5 are divisible by 5.

To verify that ***all*** numbers that end in 0 or 5 are divisible by 5 would require more than the method of verifying and modifying conjectures. When trying to show that something works for all cases, deductive reasoning should be used.

6. The following statement is an example of inductive reasoning.
All small dogs in the park today have blue collars.
Therefore, all small dogs have blue collars.
Discuss with your partner if it is possible that there existed a scenario that could have led to the conclusion. Write a short explanation.
Answers will vary. The conclusion could have been drawn if all the small dogs in the park on a given day had on blue collars.

A ***counterexample*** is a statement that disproves a conjecture.

7. Write a counterexample that would disprove "all small dogs have a blue collar."
Answers will vary. I know someone who has a small dog, and that dog does not have a blue collar.

Students often make the mistake of creating an inverse statement when trying to create a counterexample. Watch for this misconception as students work to create counterexamples throughout this and future lessons.

2.2.3 Bring It Together: Using Reasoning

Component Overview: *Students apply their learning from the prior activity to formalize the concept of using reasoning.*

2.2.3 Bring It Together: Using Reasoning

1. Determine the type of reasoning for each statement.

Statement	Type of Reasoning
The car's battery provides power to the engine, so if the battery is dead, the car won't start.	*Deductive*
I know Joe is a good cook because I've eaten at his house three times and each time the food has been amazing.	*Inductive*
The diary that is supposed to be from the 18th century might be real. It has passed a number of forensic and historical tests.	*Deductive*
The sum of two 2-digit numbers is always a 3-digit number.	*Inductive*

2. For each statement, write a counterexample that disproves the statement.

   a. If a number is multiplied by itself, then the product is even.
   $3 \cdot 3 = 9$ and 9 is not even.

   b. Adding the same number to both the numerator and the denominator of a fraction does not change the fraction's value.
   $\frac{3}{5} \neq \frac{3+2}{5+2} = \frac{5}{7}$

Use your informal observations from 2.2.2 Exploration to dictate how to summarize the learning of the Exploration. If students struggled with the concepts, then revisit the major points before beginning 2.2.3 Bring It Together. If only a few students struggled then consider using small group instruction for some while others attempt the questions independently or facilitate the component as a whole group.

Observe students as they complete the table for question 1. Their responses will provide data for what they understand and don't understand about deductive and inductive reasoning.
If you see a student with the answers inductive-deductive-inductive-deductive, it is a sign that they understand the reasoning types but have them backwards with identification.
If students have one answer incorrect, it indicates that something in the language of the statement caused confusion as to identifying the right type.
Any other combinations of incorrect answers are a sign that students may need additional remediation and support to understand the concept.

2.2.4 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

2.2.4 Wrap-Up: Cool Down

1. Write a statement that uses deductive reasoning.
   Answers will vary.

2. Write a counterexample to the statement.
   Answers will vary.